THE CATHOLIC DIOCESE OF SAME
KINDOROKO GIRLS' SECONDARY SCHOOL
FORM THREE FIRST CONTINUOUS ASSESSMENT TEST

033/1 **BIOLOGY 1**

(For Both Private and School Candidates)

Time: 2:30 Hours **Year: 2024**

Instructions

1. This paper consists of sections A, B and C with a total of **eleven (11)** questions.
2. Answer **all** questions in sections A and B and **two (2)** questions from section C.
3. Section A carries **sixteen (16)** marks, section B **fifty four (54)** marks and section C carries **thirty (30)** marks.
4. Except for drawings which must be drawn in pencil, all writings must be in **blue** or **black** ink.
5. Cellular phones and any unauthorized materials are **not** allowed in the examination room.
6. Write your **Name in Full** on every page of your answer booklet(s).

KINDOROKO BIO 1 CAT 1 2024

SECTION A (16 Marks)

Answer **all** questions in this section

1. For each of the items (i) – (x) choose the correct answer from among the given alternatives and write its letter beside the item number in the answer booklet(s) provided.

(i) Usually, forest fire occurring in conifers grows bigger and spreads quickly because: -

A Conifers are usually tall and shade the ground by their leaves
B Conifers have simple xylem and phloem hence they are very soft and light
C Conifers are evergreen with small needle-like leaves that support combustion
D Conifers contains resins that serve the oxidizing function
E Conifers have naked seeds clustered into cones hence they catch fire easily

(ii) Amanda wishes to specialize in the study of bacteria and protoctists. Therefore, she prepares to pursue: -

A Taxonomy
B Bacteriology
C Bactoprotology
D Microbiology
E Ornithology

(iii) Neilath wants to have good health. What is she supposed to do?

A Increase the intake of junky foods in her meals
B Start doing vigorous physical exercise everyday
C Take all possible measures to control stress
D Take medicines immediately when she feels unwell or tired
E Maintain proper food hygiene and have frequent sleep during the day

(iv) In the first aid procedure, which of the following items is used for removing thorns and stings left behind by bees?

A Tweezer
B Liniment
C Gentian violet
D Antiseptic
E Medical tapes

(v) Which of the following depends on cilia in order to change their position?

A Amoeba and euglena
B Paramecium and bacteria
C Ovum and paramecium
D Euglena and bacteria
E White blood cell and amoeba

(vi) Which of the following is **NOT** among characteristics of prokaryotic cells?

A They are usually flagellated
B They do not have nucleus
C They reproduce by fission
D They have no nuclear envelope
E They have relatively few organelles

(vii) Mayala was working in the laboratory. By accident she spilled unknown chemical on the floor which resulted into a permanent trace. Which of the following could be the most possible chemical that was unknown?

A Liquified petroleum gas
B Concentrated nitric acid
C Potassium permanganate solution
D Gunpowder or dynamite
E Dilute sodium hydroxide solution

(viii) Doreen and Mwanahawa were playing around a dumping place and found an organism shaped like an umbrella. Which of the following statements is true about the organism?
A It is a toadstool and belongs to the phylum Basidiomycota
B It is yeast and belongs to the phylum Ascomycota
C It is a moss plant and grows in moist, shaded and dumping places
D It is mucor and belongs to the phylum Zygomycota
E It is a mushroom and it is usually unicellular that feeds saprophytically

(ix) What does the blood flowing through the pulmonary artery have in comparison to that in the pulmonary vein?
A More oxygen and high pressure
B More oxygen and low pressure
C More carbon dioxide and high pressure
D More carbon dioxide and low pressure
E More oxygen and moderate pressure

(x) Tics are commonly found on the skin of cattle such as cows and suck blood from them. This means that:
A Ticks are preys and cattle are predators
B Ticks are ectoparasites and cattle are hosts
C Ticks are ectoparasites and cattle are endoparasites
D Ticks are heterotrophs and cattle are autotrophs
E Ticks are blood suckers and cattle are primary producers

2. Match the item in List A with the response in List B by writing the letter of the correct response besides the item number in the answer booklet(s) provided.

LIST A	LIST B
(i) A vertebral bone with a peg that aids free rotation and sideways movement of the head	A Collar bone
(ii) A tough structure that hold a muscle to a bone	B Ligament
(iii) Appendicular component that forms the forearm	C Glenoid cavity
(iv) Part of long bones responsible for blood cells synthesis	D Tendon
(v) The only long bone that lies horizontally in the human body	E Spongy bone
(vi) Articular surface of the scapula that forms a ball and socket joint with the upper arm bone	F Axis
	G Ulna bone
	H Atlas
	I Sutures
	J Humerus bone

SECTION B (54 Marks)

Answer **all** questions in this section

3. Explain six (6) consequences associated with improper disposing of wastes.

4. (a) In which ways is the bony framework important in the human body? Explain four (4) points.
(b) In three (3) points, justify the necessity of saliva in the process of food digestion.

5. (a) Suppose you are a self-employed person working as a traditional fisherwoman in your local village. Briefly explain three (3) methods you can use to preserve fish for future use.
   (b) Describe the appearance of a maize leaf by giving two (2) points. (Diagram is necessary)

6. (a) What could be the reason for:
      (i) The nose to secrete mucus
      (ii) Biology laboratory to have cages
      (iii) Drinking water to be boiled
      (iv) Stomata to be abundant on the lower surface of leaves than on the upper surface.
   (b) Malaria is one of the threatening diseases in most of African countries. Outline three (3) ways in which people can prevent themselves from this disease.

7. (a) Carefully study the equation below and then answer the questions that follow:
      Glucose ⟶ **Q** + *Carbon dioxide* + *ATP*
      (i) Give the name of the process represented by the equation above.
      (ii) Suggest the name of the product **Q** in animal muscles.
   (b) Briefly explain four (4) economic importance of the process above.

8. (a) Hamisa and Dorcas are upcoming taxonomists. During their biological activity, they classified ducks, bats and butterflies in the same group just because both have wings.
      (i) What type of classification did they apply?
      (ii) Define the type of classification named in (i) above.
   (b) The following organisms were found in an ecosystem: Lizards, grasshoppers, ferns, bacteria, frogs, butterflies and bees.
      (i) Construct a food web using the named organisms.
      (ii) From the food web above, identify two (2) organisms which are secondary consumers.
      (iii) Derive two (2) food chains from the food web constructed in (i) above.
      (iv) Why is it significant to have ferns included in this ecosystem?

SECTION C (30 Marks)

Answer **two (2)** questions in this section

9. Explain six (6) economic importance of angiosperms in the environment.

10. With examples, identify six (6) tools used in studying biology and demonstrate how they are used.

11. With the aid of a labeled diagram of the respiratory structure involved, describe the mechanism of gaseous exchange in the human body.

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